

CS 423/523 – Fall 2025



Problem Set 1

Assigned: 11 October 2025

Due: 24 October 2025 (midnight)

What to Submit:

Create a folder named `ps1_<YourLastName>` (for example, `ps1_Ates`), with the following structure and contents:

`ps<problem set number>_<problem number>.py` → The main script to run.

`/input/` → Directory containing input images, videos or other data used

`/output/` → Directory containing output image, videos or other data generated

`ps<problem set number>_report.pdf` → A PDF report file. (Include figures, discussions, methods, etc., but do not include any code.)

Zip your folder and submit on **LMS**. Submissions that do not obey the guideline above will **NOT** be evaluated.

Problem 1: (Basic image operations)

Download a color image to your computer.

(a) Write a Python script to resize the image to 75% of its original dimensions. Save the resized image as a PNG file.

(b) Convert the image to grayscale and apply a Gaussian blur with a sigma value of 2. Display the blurred grayscale image.

(c) Apply edge detection using the Canny algorithm to the grayscale image. Display the edges found in the image.

(d) Create a binary mask from the edge-detected image (edges are white, and the rest is black). Apply this mask to the original color image, making the edges prominent while dimming the rest of the image. Display the result.

(e) Perform histogram equalization on the grayscale image to enhance its contrast. Display the histogram-equalized image.

Problem 2: (Video Processing with Parameterized Edge Enhancement)

Write a Python script to capture a live video stream from your computer's webcam (If it is unavailable, use arbitrary video of your choice). For each frame:

- (a) Convert the frame to grayscale.
- (b) Apply edge detection using the Canny algorithm to the grayscale frame.
- (c) Create a binary mask from the edge-detected frame.
- (d) Apply the mask to the original frame, making the edges prominent while dimming the rest.
- (e) Enhance the edges in the original frame by adding the edge-detected frame back to the original frame, controlled by an edge enhancement factor. This factor determines the intensity of edge enhancement: a value greater than 1 enhances the edges, while a value of 1 leaves the original frame unchanged.
- (f) Display the original frame, the masked frame, and the edge-enhanced frame by concatenating them side by side, with their names ("Original", "Masked", and "Edge-Enhanced").
- (g) Display the current fps on the concatenated frames. The video stream should stop when you hit "q" on your keyboard.

Hint: Use the `cv2.addWeighted` function in Python to blend the original frame with the edge-detected frame. The syntax is `cv2.addWeighted(src1, alpha, src2, beta, gamma)`, where `src1` and `src2` are the images to blend, `alpha` and `beta` are the weights for each image, and `gamma` is a scalar added to the result. In our case, you can use it like `cv2.addWeighted(frame, 1, cv2.cvtColor(mask, cv2.COLOR_GRAY2BGR), edge_enhancement_factor, 0)`. Play with the `edge_enhancement_factor` and observe the results.